Math Quiz 1

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1 Problem 1

D is defined in terms of y, so the equation for this parabola has to be:

$$y - c = r(x - b)^2$$

We are given the focus and directrix

$$F : (x, y) = (2, -1),\ D : y = 3$$

And the equation's focus and directrix shifts are

$$F = \left(b, c + \frac{1}{4r}\right),\ D : y = c - \frac{1}{4r}$$

Plug in relevant values from given numbers into equation's F and D

$$2 = b,\ -1 = c + \frac{1}{4r},\ 3 = c - \frac{1}{4r}$$

Make values equal by adding 4 to equation that was equal to -1

$$3 = c + \frac{1}{4r} + 4$$

System of equations to solve for r

$$c + \frac{1}{4r} + 4 = c - \frac{1}{4r}$$

$$\frac{2}{4r} = -4$$

$$\frac{1}{2r} = -4$$

$$1 = -8r$$

$$r = -\frac{1}{8}$$

Plug in r to solve for c

$$D : 3 = c - \frac{1}{4(-\frac{1}{8})} \rightarrow 3 = c - -\frac{1}{1/2}$$

$$3 = c + 2$$

$$c = 1$$

Have everything to complete equation giving us a final answer of:

$$y - 1 = -\frac{1}{8}(x - 2)^2$$

2 Problem 2

The inequality can also be expressed as

$$f_\alpha(4) > f_{10}(10)$$

$\alpha > 0$ for all observed instances of 4^α
This means that since $f_\alpha(10)$ where $\alpha = 10$ that both functions are **Strictly increasing** since $\alpha > 0$ is always true

a) ~~Not large enough, if we use the laws of exponents, we can multiply both sides by 10th root~~

$$\cancel{4^{10*\frac{1}{10}} > 10^{10*\frac{1}{10}}}$$

~~Which gives us a false inequality~~

$$\cancel{4 > 10}$$

REVISED A
If $n = 10$, then inequality is

$$4^{10} > 10^{10}$$

Since both sides are increasing, and to the same power, we can compare bases directly

$$4 > 10$$

Which is an false inequality, meaning $n = 10$ isn't enough

b) Yes it's large enough, we can seperate 4^{20} by using law of exponents

$$4^{20} = (4^{10})^2$$

$$(4^{10})^2 = (4^2)^{10}$$

$$(4^2)^{10} = 16^{10}$$

~~Apply same principle as in a, but being careful to distribute $\frac{1}{10}$ power equally~~

$$~~4 * 4 > 10~~$$

Now that we have both sides of inequality to same power, we can again compare bases since both are also increasing

$$16^{10} > 10^{10} \rightarrow 16 > 10$$

Inequality is true, so n=20 is enough

c) ~~No it isn't. By breaking up both sides of equation into 5th degree exponents, we can prove 4^{15} isn't big enough while making sure the bases and exponents stay as *whole numbers*~~

$$~~4^5 * 4^5 * 4^5 = 10^5 * 10^5~~$$

~~Multiply both sides by $\frac{1}{5}$ power, making sure to distribute correctly~~

$$~~4^3 > 10^2~~$$

$n = 15$ isn't enough. By applying same principal as other two, we are going to use laws of exponents to get get both sides to a common exponent

$$4^{15} > 10^{10}$$

This time have to rearrange both sides, this time using 5 since it's a common multiple between 15 and 10

$$(4^5)^3 > (10^5)^2$$

$$(4^3)^5 > (10^2)^5$$

$$64^5 > 100^5$$

Now if we compare bases after getting to common exponent when both are strictly increasing, we get:

$$64 > 100$$

This inequality is false, so $n = 15$ isn't enough

3 Problem 3

a) $4x^3$, previous examples follow a nx^{n-1} pattern, but we haven't proved this true at time of the quiz (but my future self correcting this has now seen it proven)

b) Any value s in relation to a fixed value x can be compared vertically by

$$s^4 - x^4$$

Unlike above (when compared to ones for previous degrees), the tangent line equation doesn't change

$$m(s-x)$$

We want the line to always be below the parabola, so for any value s, we want

$$s^4 - x^4 \geq m(s-x)$$

Move over $m(s-x)$ and factor out common $(s-x)$ gives us

$$(s^3 + s^2x + sx^2 + x^3 - m)(s-x) \geq 0$$

Have to "know" for geometric argument, fortunately, we do know

$$m = 4x^3$$

$$(s^3 + s^2x + sx^2 - 3x^3)(s-x) \geq 0$$

~~Parts of expression cancel out if multiplied by$(s-x)$~~

~~$(s^4 - 4x^3s + 3x^4) \geq 0$~~

Instead of above, will realize that $s^3 + s^2x + sx^2 - 3x^3$ CAN be factored by $(s-x)$.

$$(s-x)(s^3 + s^2x + sx^2 - 3x^3) = (s-x)(\mathbf{s-x})(\mathbf{s^2 + 2sx + 3x^2})$$

Fact check that $(s-x)$ is a factor

$$(s-x)(s^2 + 2sx + 3x^2) = (s^3 + 2s^2x + 3x^2s) + (-s^2x - 2x^2s - 3x^3)$$

$$s^3 + (2s^2x - s^2x) + (3x^2s - 2x^2s) - 3x^3 = s^3 + s^2x + x^2s - 3x^3$$

Math above checks out, so we carry on
Can factor even further since $s^2 + 2sx + 3x^2$ can be factored again

$$s^2 + 2sx - 3x^2 = (s+x)(s+x) + 2x^2$$

After all this factoring we are left with

$$(s-x)(s-x)(s+x)(s+x) + 2x^2 = (s-x)^2(s+x)^2 + 2x^2$$

Every factor is to 2nd power, so are **all** are non-negative, meaning:

$$(s-x)^2(s+x)^2 + 2x^2 \geq 0$$

Which proves $4x^3 = m$

~~I couldn't figure out where to go next in my 4 thinking hours. s^4 and $3x^4$ are always non negative, but $-4x^3s$ can be anything. Because of that, I wasn't sure how I would prove that the inequality is always greater than or equal to 0. If the shaky limits argument was applied to this as it was, it would be $x^4 - 4x^4 + 3x^4$ which is just zero~~

4 Problem 4

For fixed points $P(a, a^2)$ and $Q(b, b^2)$, we can connect them to an unfixed point $R(c, c^2)$ between them somewhere on the parabola
$\tilde{R}$ is a possible Point R where the triangle from the 3 points will have the greatest area of any in the parabola
The point $\tilde{R}$ is located exactly between P and Q represented by $\tilde{R} = \frac{a+b}{2}$
$\tilde{R}$ can equal R, but not usually
The slope of the tangent line of $\tilde{R} = a + b$ and it is parallel to Line PQ
For all triangles:

$$A_{tri}PQR \leq A_{tri}P\tilde{R}Q$$

We can replace A_{tri} with actual formula for area of a triangle to compare which variables control it
Let $\tilde{H}$ represent height of the triangle with $\tilde{R}$

$$\frac{PQ * H}{2} \leq \frac{PQ * \tilde{H}}{2}$$

after canceling out constants:

$$H \leq \tilde{H}$$

the only variable that matter for the area of a triangle in the parabola is height
$\tilde{R}$ is also the only point on the parabola that is also on tangent $\overleftrightarrow{L}$
This makes sense given that tangent lines also intersect what they are tangent to *once* (in the general area, which in case of 2nd degree parabola is everwhere since it won't meet the parabola *ever* again)

Otherwise, the line is below the parabola

$\overleftrightarrow{L} \parallel \overleftrightarrow{PQ}$ since they have the same slope

Using these premises, and knowing that height is the only indicator of A_{tri}, we can say about height:

H = Distance from $\overleftrightarrow{PQ}$ to point R
Additionally, the distance between $\overleftrightarrow{PQ}$ and $\overleftrightarrow{L}$ = Distance from $\overleftrightarrow{PQ}$ to point R (which equals H) + distance from point R to $\overleftrightarrow{L}$
However, $\tilde{R}$ is on $\overleftrightarrow{L}$, so the distance from point $\tilde{R}$ to $\overleftrightarrow{L}$ is 0
This means the distance between $\overleftrightarrow{PQ}$ and $\overleftrightarrow{L} = H + 0$, meaning the distance between the parallel lines equals H for, and only for $\tilde{R}$
This is also the greatest height since the maximum height is the distance between $\overleftrightarrow{PQ}$ and $\overleftrightarrow{L}$, which H is when solved for $\tilde{R}$
Since height is the only relevant variable to the area of the triangle, the greatest height at $\tilde{R}$ is also the greatest area, proving why Archimedes chose $\tilde{R}$ for the 3rd point

~~I will end my write up on 4 here since I ran out of time and was likely not going to progress further in the problem if I spent more time. Had thought in head of using the parallel lines of PQ and tangent line $\tilde{R}$ to create a rectangle, but ultimately didn't see how that would allow me to find relationship between area.~~